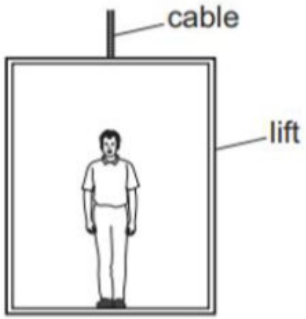
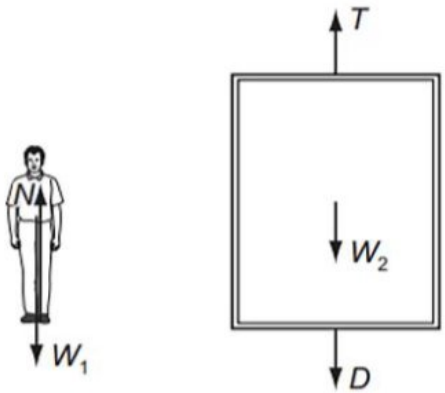
4 The diagram shows a man standing in a lift.



The forces acting on the man and the forces acting on the lift are shown below.



N is the force from the lift floor on the man.
W_1 is the weight of the man.
T is the tension in the lift cable.
W_2 is the weight of the lift.
D is the force from the man on the lift floor.

Which of the following statements is correct?

A $(W_1 + W_2)$ is always equal to T.

B If $N = W_1$, the lift must be at rest.

C N and W_1 are always equal and opposite.

D If $T = (D + W_2)$, the lift must have a constant velocity.